

Frankfurt Municipal District Heating Waste Heat Export Power Purchase Agreement

Bulletin ID: PPA-FFM-HEAT-2025-01 | **Classification:** LEGAL_COMMERCIAL | **Published:** 2025-04-01

Issuing Agency / Entity: Mainova AG (Frankfurt Municipal Utility) & Aethelgard DC-1

Section 1: Situation Assessment & Incident Summary

Binding energy partnership under the German Energy Efficiency Act (EnEfG). Aethelgard DC-1 commits to exporting continuous waste thermal energy into the Frankfurt municipal district heating network.

Section 2: Impacted Trade Routes & Hardware Components

Primary Coolant Loop Pump B (SKU-PUMP-HALL1-B), Kelvion heat exchangers, and secondary Loop-A liquid cooling loops.

Section 3: Supply Chain Lead Time & Financial Forecast

Minimum continuous thermal export target: 12.5 MW thermal at minimum 58°C supply temperature.

Section 4: Contractual & Municipal Penalties

Under Section 8.4 (Thermal Export Default), if an unexcused cooling loop shutdown causes thermal supply to fall below 60% of baseline for more than 4 consecutive hours, Aethelgard shall pay liquidated damages of €45,000 per 24-hour period to Mainova AG.

Section 5: Recommended Mitigation Actions

Maintain redundant N+1 cooling pump failover and backup heat dump radiators.